

DESKTOP SSDs

NVMe is now mainstream

A year back, getting an NVMe drive wasn't feasible because that one NVMe drive would cost more than a traditional SSD coupled with a 2-4 TB hard drive. On top of that, you'd have to get a motherboard that supported x4 PCIe lanes on the M.2 ports, which again, wasn't cheap. This year, we've seen a significant drop in NVMe SSD prices and the M.2 interface has become enormously popular, much so that even budget boards have an M.2 slot or two but not all have x4 PCIe lanes. Aside from NVMe, another key development is with the silicon, and that's the 3D XPoint technology. As of now it's just Intel that has consumer drives with 3D XPoint technology which they market under the Optane brand and we don't know if third-party manufacturers will be licensed to make additional SKUs. We're yet to get our hands on any of the high-capacity Optane SSDs so we can't say how well they perform compared to the competition, but if preliminary reports are anything to go by, then Optane drives easily outperform M.2 NVMe SSDs. For now, we're stuck with M.2 NVMe SSDs until NVDIMMs products are unveiled. For the uninitiated, NVDIMMs are Non-Volatile Dual Inline Memory Modules, they're basically RAM modules which don't lose data upon losing power.



Corsair MP500
Price: ₹16,499



Kingston KC1000
Price: ₹22,531



WD Blue
Price: ₹14,771



Samsung 960 Pro
Price: ₹45,682



2017 ZERO1 AWARD WINNER

Samsung 960 Pro

It's no wonder that the Samsung 960 Pro gets the Zero1 award given that the performance has been way ahead of everything else. Samsung has the advantage of owning fabs which allows them to maintain a good level of control over the manufacturing process of both, the Polaris controller and the 3D V-NAND. When it comes to examining the different SSD controllers, most perform well in synthetic benchmarks but in real world tests and trace-based analysis using IOMeter, they falter often. With the Polaris controller, it maintains a consistent lead over let's say, a Phison or Marvell controller. Moreover, the Samsung 960 Pro has a more consistent performance post conditioning which is where most other SSDs start to crumble.

RUNNER UP

Corsair MP500 & Kingston KC1000

The Kingston KC1000 is Kingston's first ever NVMe SSD and boy does it perform good. Like the Samsung 960 Pro, the Kingston KC1000 is a consistent performer, though it peaks at a slightly lower performance level as compared to the 960 Pro. Despite having a Phison controller, the KC1000 performs not only in sequential transfers of large file sizes but even with small files across varying queue depths. Similarly, the Corsair MP500 has spectacular performance since it too is based on the same Phison E7 controller. We ended up getting a slightly lower performance on the Corsair MP500 than the Kingston KC1000. The review unit for the MP500 was a 240 GB SKU while the KC1000 was a 480 GB SKU. We suspect the higher capacity drive helped the Kingston SKU since there are more NAND chips for simultaneous transfers.

BEST BUY

WD Blue

WD Blue might not have scored high in our benchmarks like the WD Black NVMe SSD but being one of the more economically priced SKUs, the WD Blue offers you more GB/Rupee. It's performance isn't like other NVMe drives since its transfer rates are around 560 MB/s which is what most SATA SSDs end up with. So essentially, you're getting SATA SSD speeds on the M.2 interface. Given that NVMe SSDs have transfer speeds going up to 3,500 MBps, the WD Blue appears to be on the other end of the spectrum. However, if you want to upgrade a thin form-factor laptop, then the WD Blue is a great unit to start with. 